

Statistical Analysis of Bernoulli Sets: Distributions, Confidence Intervals, and Performance Measures

Alexander Towell
atowell@siue.edu

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Abstract

While the latent/observed framework provides conceptual clarity for Bernoulli types, practical applications require understanding the statistical properties of these observations. We present a comprehensive analysis of the distributions that arise when observing latent sets through noisy channels. For finite sets, the number of false positives and false negatives follow binomial distributions, leading to beta-distributed error rates. We derive asymptotic normal approximations, confidence intervals, and the distributions of key performance measures including positive predictive value (precision). Additionally, we develop interval arithmetic methods for propagating uncertainty when error rates themselves are uncertain. These results enable practitioners to quantify uncertainty, design systems with statistical guarantees, and analyze the performance of probabilistic data structures in real-world applications.

1 Introduction

The Bernoulli framework distinguishes between latent mathematical objects and their computational observations. While this provides theoretical elegance, practical applications demand answers to statistical questions:

- If we observe a set with nominal false positive rate $\alpha = 0.01$, what is the actual false positive rate likely to be?
- How many false positives should we expect when querying 1 million elements?
- What confidence intervals can we place on precision and recall?
- How do we handle uncertainty in the error rates themselves?

This paper provides rigorous statistical analysis of Bernoulli sets, focusing on finite-sample distributions and practical tools for uncertainty quantification.

2 Fundamental Distributions

2.1 The Statistical Model

We model the observation of a latent set $S \subseteq U$ through a noisy channel that produces an observed set $\tilde{S}$. The key insight is that for finite sets, the observation errors follow well-understood probability distributions.

Axiom 1 (Independence of Errors). *Each element's observation error is independent:*

$$\mathbb{P} \left[\mathbf{1}_{\tilde{S}}(x) \neq \mathbf{1}_S(x) \mid \mathbf{1}_{\tilde{S}}(y) \neq \mathbf{1}_S(y) \right] = \mathbb{P} \left[\mathbf{1}_{\tilde{S}}(x) \neq \mathbf{1}_S(x) \right] \quad (1)$$

for all distinct $x, y \in U$.

This independence assumption, while idealized, holds for many practical implementations including Bloom filters and hash-based data structures.

2.2 Distribution of False Positives

Theorem 2.1 (False Positive Distribution). *Given a latent set S with $n = |\bar{S}|$ negative elements (elements not in S), the number of false positives when observing through a channel with false positive rate α follows:*

$$\text{FP}_n \sim \text{Binomial}(n, \alpha) \quad (2)$$

with expectation $\mathbb{E}[\text{FP}_n] = n\alpha$ and variance $\text{Var}[\text{FP}_n] = n\alpha(1 - \alpha)$.

Proof. Each of the n elements in $\bar{S}$ has probability α of being incorrectly observed as belonging to $\tilde{S}$. By the independence axiom, these are n independent Bernoulli trials. $\square$

Corollary 2.2 (Observed False Positive Rate). *The observed false positive rate $\text{FPR}_n = \text{FP}_n/n$ has:*

- *Expectation:* $\mathbb{E}[\text{FPR}_n] = \alpha$
- *Variance:* $\text{Var}[\text{FPR}_n] = \alpha(1 - \alpha)/n$
- *Distribution:* Scaled binomial on support $\{0, 1/n, 2/n, \dots, 1\}$

2.3 Distribution of False Negatives

By symmetry, false negatives follow an analogous distribution:

Theorem 2.3 (False Negative Distribution). *Given a latent set S with $p = |S|$ positive elements, the number of false negatives follows:*

$$\text{FN}_p \sim \text{Binomial}(p, \beta) \quad (3)$$

2.4 Joint Distribution of Classification Outcomes

For a complete statistical picture, we need the joint distribution of all classification outcomes:

Theorem 2.4 (Joint Distribution). *Given p positive and n negative elements in the latent set, the joint distribution of classification outcomes is:*

$$(\text{TP}_p, \text{FN}_p, \text{FP}_n, \text{TN}_n) \sim \text{Product of Independent Binomials} \quad (4)$$

$$\text{TP}_p \sim \text{Binomial}(p, 1 - \beta) \quad (5)$$

$$\text{FN}_p \sim \text{Binomial}(p, \beta) \quad (6)$$

$$\text{FP}_n \sim \text{Binomial}(n, \alpha) \quad (7)$$

$$\text{TN}_n \sim \text{Binomial}(n, 1 - \alpha) \quad (8)$$

3 Asymptotic Approximations

For large sets, normal approximations provide computational efficiency and theoretical insight.

3.1 Central Limit Theorem for Error Rates

Theorem 3.1 (Asymptotic Normality). *As $n \rightarrow \infty$, the observed false positive rate converges in distribution:*

$$\sqrt{n}(\text{FPR}_n - \alpha) \xrightarrow{d} \mathcal{N}(0, \alpha(1 - \alpha)) \quad (9)$$

Equivalently:

$$\text{FPR}_n \approx \mathcal{N}\left(\alpha, \frac{\alpha(1 - \alpha)}{n}\right) \quad (10)$$

Proof. By the Central Limit Theorem applied to the mean of n independent Bernoulli(α) random variables. $\square$

3.2 Confidence Intervals

Corollary 3.2 (Asymptotic Confidence Intervals). *A $(1 - \alpha)$ confidence interval for the observed false positive rate is:*

$$\alpha \pm z_{\alpha/2} \sqrt{\frac{\alpha(1 - \alpha)}{n}} \quad (11)$$

where $z_{\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the standard normal distribution.

Example 3.3 (Bloom Filter Confidence). *A Bloom filter with design false positive rate $\alpha = 0.01$ and $n = 10^6$ negative elements has 95% confidence interval:*

$$0.01 \pm 1.96 \sqrt{\frac{0.01 \times 0.99}{10^6}} = [0.0098, 0.0102] \quad (12)$$

The observed rate will be within 2% of the nominal rate with high probability.

4 Performance Measure Distributions

4.1 Positive Predictive Value (Precision)

The positive predictive value is the probability that an element observed in $\tilde{S}$ is actually in the latent set S .

Definition 4.1 (PPV as Random Variable). *Given p positives and n negatives:*

$$\text{PPV} = \frac{\text{TP}_p}{\text{TP}_p + \text{FP}_n} \quad (13)$$

where TP_p and FP_n are independent binomial random variables.

Theorem 4.2 (Expected PPV). *The expected positive predictive value is approximately:*

$$\mathbb{E}[\text{PPV}] \approx \frac{\bar{t}_p}{\bar{t}_p + \bar{f}_p} + \frac{\bar{t}_p \sigma_{\bar{f}_p}^2 - \bar{f}_p \sigma_{\bar{t}_p}^2}{(\bar{t}_p + \bar{f}_p)^3} \quad (14)$$

where:

$$\bar{t}_p = p(1 - \beta) \quad (\text{expected true positives}) \quad (15)$$

$$\bar{f}_p = n\alpha \quad (\text{expected false positives}) \quad (16)$$

$$\sigma_{t_p}^2 = p\beta(1 - \beta) \quad (\text{variance of true positives}) \quad (17)$$

$$\sigma_{f_p}^2 = n\alpha(1 - \alpha) \quad (\text{variance of false positives}) \quad (18)$$

Proof Sketch. Using the delta method for the ratio of random variables, we expand $g(X, Y) = X/(X + Y)$ around $(\mathbb{E}[X], \mathbb{E}[Y])$ to second order. $\square$

Remark 4.3 (Interpretation). *The first term is the naive expectation treating denominators as fixed. The correction term accounts for the variance in both true and false positives, showing that higher variance in false positives improves expected precision while higher variance in true positives degrades it.*

4.2 Distribution Shape of PPV

While the exact distribution of PPV is complex, we can characterize its behavior:

Proposition 4.4 (PPV Concentration). *In the high-precision regime where $\bar{f}_p \ll \bar{t}_p$:*

$$\text{Var}[\text{PPV}] \approx \frac{\sigma_{f_p}^2}{\bar{t}_p^2} = \frac{n\alpha(1 - \alpha)}{[p(1 - \beta)]^2} \quad (19)$$

The variance is dominated by fluctuations in false positives.

5 Uncertain Error Rates

In practice, we often have uncertainty about the error rates themselves. This section develops interval arithmetic methods for propagating this uncertainty.

5.1 Interval Representations

Definition 5.1 (Interval Error Rates). *When error rates are uncertain, we represent them as intervals:*

$$[\alpha] = [\underline{\alpha}, \bar{\alpha}] \quad (20)$$

$$[\beta] = [\underline{\beta}, \bar{\beta}] \quad (21)$$

where $\underline{\alpha}$ and $\bar{\alpha}$ are lower and upper bounds respectively.

5.2 Performance Measures with Interval Arithmetic

Theorem 5.2 (Interval Accuracy). *Given interval error rates and proportion of positives $\lambda \in [\underline{\lambda}, \bar{\lambda}]$, the accuracy lies in:*

$$\text{Accuracy} \in \left[\min_{\substack{\alpha \in [\alpha] \\ \beta \in [\beta] \\ \lambda \in [\lambda]}} \text{Acc}(\alpha, \beta, \lambda), \max_{\substack{\alpha \in [\alpha] \\ \beta \in [\beta] \\ \lambda \in [\lambda]}} \text{Acc}(\alpha, \beta, \lambda) \right] \quad (22)$$

Example 5.3 (Worst-Case Analysis). *With complete uncertainty about the proportion of positives ($[\lambda] = [0, 1]$) and known false positive rate α :*

$$\text{Accuracy} \in [1 - \alpha, 1] \quad (23)$$

The worst case occurs when all elements are negative and we have maximum false positives.

5.3 Confidence Intervals vs. Interval Arithmetic

It's crucial to distinguish two types of intervals:

- **Confidence intervals:** Probabilistic statements about where a random quantity likely falls
- **Interval arithmetic:** Deterministic bounds when parameters are known to lie in ranges

These can be combined:

Proposition 5.4 (Combined Uncertainty). *If the true false positive rate lies in $[\alpha] = [0.01, 0.02]$ and we have $n = 10^6$ negatives, then with 95% confidence, the observed false positive count lies in:*

$$\text{FP}_n \in [9, 800 \pm 196, 20, 000 \pm 280] = [9, 604, 20, 280] \quad (24)$$

6 Applications

6.1 Bloom Filter Analysis

Example 6.1 (Bloom Filter Precision). *A Bloom filter with m bits, k hash functions, and n stored elements has:*

- *Expected FPR: $\alpha = (1 - e^{-kn/m})^k$*
- *For a query workload with proportion λ of positive queries:*

$$\mathbb{E}[\text{PPV}] \approx \frac{\lambda}{\lambda + (1 - \lambda)\alpha} \quad (25)$$

- *With n_q queries, the observed PPV has approximate variance:*

$$\text{Var}[\text{PPV}] \approx \frac{\lambda(1 - \lambda)\alpha(1 - \alpha)}{n_q[\lambda + (1 - \lambda)\alpha]^2} \quad (26)$$

6.2 Distributed Set Operations

Example 6.2 (Union with Uncertain Sources). *When computing $\tilde{A} \cup \tilde{B}$ where sets come from different sources with error rates in intervals:*

- *A observed with $\alpha_A \in [0.01, 0.02]$*
- *B observed with $\alpha_B \in [0.005, 0.015]$*
- *Union false positive rate: $\alpha_{A \cup B} \in [0.01495, 0.0347]$*

The interval widens due to uncertainty composition.

6.3 Privacy-Preserving Queries

Example 6.3 (Differential Privacy via Bernoulli Noise). *Adding Bernoulli noise for privacy:*

- *True answer: Set S with $|S| = 1000$*
- *Add false positives with rate $\alpha = 0.1$*
- *Remove true positives with rate $\beta = 0.1$*
- *Expected observed size: $900 + 0.1 \times |\bar{S}|$*
- *Privacy parameter: $\epsilon = \log(0.9/0.1) = 2.2$*

7 Advanced Topics

7.1 Higher-Order Moments

Beyond expectation and variance, higher moments provide additional insight:

Proposition 7.1 (Skewness of FPR). *The observed false positive rate has skewness:*

$$\text{Skew}(\text{FPR}_n) = \frac{1 - 2\alpha}{\sqrt{n\alpha(1 - \alpha)}} \quad (27)$$

For small α , the distribution is right-skewed, with occasional large deviations above the mean.

7.2 Multivariate Analysis

When analyzing multiple sets simultaneously:

Theorem 7.2 (Joint Distribution of Multiple Sets). *For k independent observed sets of the same latent set S :*

$$(\text{FPR}_1, \dots, \text{FPR}_k) \approx \text{Multivariate Normal}(\alpha \mathbf{1}, \frac{\alpha(1 - \alpha)}{n} \mathbf{I}) \quad (28)$$

7.3 Sequential Testing

When observations arrive sequentially:

Proposition 7.3 (Sequential Confidence Intervals). *After observing t elements with f_t false positives, a Bayesian credible interval for α using a uniform prior is:*

$$\alpha \in \left[\frac{f_t + 1}{t + 2}, \frac{f_t + 1}{t + 2} \right]_{95\%} \quad (29)$$

based on the $\text{Beta}(f_t + 1, t - f_t + 1)$ posterior.

8 Practical Guidelines

8.1 Sample Size Requirements

To achieve relative error ϵ in estimating false positive rate with confidence $1 - \alpha$:

$$n \geq \frac{z_{\alpha/2}^2(1 - \alpha)}{\epsilon^2 \alpha} \quad (30)$$

Example 8.1. *To estimate $\alpha = 0.01$ within 10% relative error with 95% confidence:*

$$n \geq \frac{1.96^2 \times 0.99}{0.1^2 \times 0.01} = 38,032 \quad (31)$$

8.2 Choosing Between Models

- **Use exact binomial:** When $n < 1000$ or $n\alpha < 10$
- **Use normal approximation:** When $n\alpha > 10$ and $n(1 - \alpha) > 10$
- **Use Poisson approximation:** When $\alpha < 0.01$ and $n\alpha < 20$

8.3 Cardinality Estimation

In many applications, we need to estimate the cardinality of the latent set S given only observations of $\tilde{S}$:

Theorem 8.2 (Method of Moments Estimator). *For finite universe U with $|U| = u$, given $\tilde{S}$ with error rates α and β :*

$$|\hat{S}| = \frac{|\tilde{S}| - \alpha u}{(1 - \beta) - \alpha} \quad (32)$$

Proof. The expected size of $\tilde{S}$ is:

$$\mathbb{E}[|\tilde{S}|] = |S|(1 - \beta) + (u - |S|)\alpha \quad (33)$$

Solving for $|S|$ yields the estimator. □

For data structures with known space complexity:

Proposition 8.3 (Space-Based Estimator). *If a data structure uses b expected bits per element, then:*

$$|\hat{S}| = \frac{\text{BitLength}(\tilde{S})}{b} \quad (34)$$

For optimal structures, $b = -\log_2 \alpha$ bits per element.

Example 8.4 (Bloom Filter Cardinality). *A Bloom filter with m bits and k hash functions storing approximately n elements has:*

$$\hat{n} = -\frac{m}{k} \ln \left(1 - \frac{\text{bits set}}{m} \right) \quad (35)$$

9 Related Work

This statistical analysis builds on:

- Classical work on Bloom filter analysis [1?]
- Probabilistic data structure theory [?]
- Interval arithmetic methods [?]
- Statistical analysis of streaming algorithms [2]

10 Conclusions

While the latent/observed framework provides conceptual elegance, practical applications demand rigorous statistical analysis. We have shown that:

- Observation errors in finite sets follow binomial distributions
- Asymptotic normality enables efficient confidence interval computation
- Performance measures like PPV have complex but analyzable distributions
- Interval arithmetic provides tools for worst-case analysis
- Combined probabilistic and interval methods handle multiple uncertainty sources

These results enable practitioners to:

- Design systems with statistical guarantees
- Quantify uncertainty in probabilistic data structures
- Make informed space-accuracy trade-offs
- Validate implementations against theoretical predictions

The statistical view complements the algebraic view of Bernoulli types, providing the quantitative tools necessary for real-world deployment.

References

- [1] Burton H Bloom. Space/time trade-offs in hash coding with allowable errors. *Communications of the ACM*, 13(7):422–426, 1970.
- [2] Graham Cormode and S Muthukrishnan. An improved data stream summary: the count-min sketch and its applications. In *Journal of Algorithms*, volume 55, pages 58–75. Elsevier, 2005.